MCO-1D

Software Reference — Rev 24

MCO-1D ornithopter SAR drone — September 2026

Document	MCO-1D_Software_Reference_v24
Revision	September 2026
New in this revision	All new GS commands · 6 genome axes · 30 MAP-Elites bins · 97 hypotheses · soaring variometer · AI observer · fl
Audience	Commissioning engineers

Architecture

75,098 lines · 97 hypotheses · 50 SG tasks · 338 GS commands · 6 genome axes · 30 MAP-Elites intent bins · 20 ALSS Class B segments · 35 ALSS Class C segments · 28 discovery parameter bounds. 50 Hz main loop: pilot_intent → CIC → MIC → wing_guard → stroke_smoothing (+ soaring_vario) → position_hold (+ flight_monitor + AI_observer @1Hz).

Reference Bugs

`vsi_ref [fpm]`
Set VSI bug. Display only.

`vsi_ref_clear`
Clear VSI bug.

`hdg_ref [0-360]`
Set heading bug on DI.

`hdg_ref_clear`
Clear heading bug.

`alt_ref [alt] [ft|m]`
Set altitude bug on altimeter.

`alt_ref_clear`
Clear altitude bug.

`precision_status`
All three bugs + trim quality.

Flight Monitor — NEW v24

- `monitor_report`
  Full flight monitor report. All tiers including silent log. Alerts, trends, trim baseline, rope rate, oscillation state.
- `monitor_enable`
  Enable flight monitor (default on).
- `monitor_disable`
  Silence all alerts. Not recommended.
- `monitor_reset`
  Clear monitor state and history.
- `monitor_thresholds`
  Show all threshold values. Tier 1: bat<19V, pwr>4.5kW, vib>0.35g, guard>10/min, osc>3 in 2s. Tier 2: bat<21.5V, pwr>3.5kW, vib>0.15g, trim>50fpm, rope>2%/min. Tier 3: log only — learning plateau, trim baseline.

AI Observer — NEW v24

- `observer_approve`
  Approve pending proposal. Executes real learning system write: cascade bounds → _exploration_state, genome boost → _MW_*, archive commit → _me_archive.
- `observer_decline`
  Decline pending proposal.
- `observer_report`
  Full observer report: Ps trend, archive coverage %, cascade clustering, genome weights all 6 axes, recent proposals + actions.
- `observer_status`
  Brief status: active, pending proposal.
- `observer_enable`
  Enable AI observer.
- `observer_disable`
  Disable AI observer (still watches, does not propose).

Soaring / Motor Glider — NEW v24

- `vario_mode [basic|te|netto]`
  Select variometer display mode. basic: stroke-smoothed VSI. te: total energy (TE = VSI + V×dV/dt / g). netto: air mass movement only (netto = TE – polar_sink(V)). Soaring mode auto-activates when lever < 15%.
- `soaring_status`
  Full soaring status: variometer readings (basic/TE/netto), thermal detection, thermal strength, MacCready advisory, polar performance, refined polar if available.

Discovery / Optimizer / Genome — NEW v24

- `disc_registry`
  All 97 hypotheses with queue position, target, priority, evaluation count, best Ps. Shows ALSS B/C segment counts and discovery bound count.
- `disc_mode_focus_list`
  All 15 mission foci with matching hypothesis targets and count.
- `discovery_hypothesis_select`
  Select next hypothesis from priority queue. Respects focus routing and 5-minute per-hypothesis cooldown.
- `mechanism_weight_report`
  All 6 genome axes: roll(5), pitch(5), yaw(5), stability(5), soaring(5), glide(4). Bar chart of weights per mechanism.

Genome Axes — v24

`_MW_ROLL`
roll_amp, roll_sweep, roll_aoa, roll_phase, roll_dih

`_MW_PITCH`
pitch_freq, pitch_aoa, pitch_sweep, pitch_phase, pitch_dih

`_MW_YAW`
yaw_phase, yaw_dih, yaw_amp, yaw_sweep, yaw_h26

- `_MW_STABILITY [NEW]`
  geom_dihedral, stroke_dihedral, weathercock_casc, proverse_yaw, speed_adapt_dih
- `_MW_SOARING [NEW]`
  glide_aoa, glide_sweep, slow_flap_assist, thermal_circle, maccready_speed
- `_MW_GLIDE [NEW]`
  min_sink_aoa, best_glide_aoa, span_extension, drag_minimise

MAP-Elites Niche Bins (30 total)

`Original 22`
hover, cruise, sar, storm, sprint, roll, turn, brake, approach, clap, antiphase, fast_cruise, top_speed, climb_rate, climb_angle, aileron_roll, snap_roll, barrel_roll, chandelle, immelmann, precision, water_approach

- `New 8 [NEW]`
  stability, soaring, glide, thermal, density_alt, level_body, acoustic, sensor_point

New Telemetry Keys — v24

- `te_vario_ms / te_vario_fpm`
  Total energy variometer reading.

- `netto_vario_ms / netto_vario_fpm`
  Netto variometer — air mass only.
- `thermal_detected`
  True if netto > 0.2 m/s for >5s.
- `thermal_str_ms`
  Smoothed thermal strength (m/s).
- `maccready_V_ms / maccready_V_kt`
  MacCready optimal inter-thermal speed.
- `polar_sink_ms`
  Polar predicted sink at current V.
- `sor_active`
  Soaring mode active (lever < 15%).
- `vario_mode`
  Current vario mode: basic/te/netto.
- `mon_active`
  Flight monitor active.
- `mon_flight_time_s`
  Flight time since arm (seconds).
- `mon_alert_count`
  Total alerts this session.
- `mon_oscillating`
  True if VSI oscillation detected.
- `mon_trim_vsi_fpm`
  Baseline VSI at neutral controls.
- `obs_messages`
  List of observer messages this tick.

`hdg_ref_deg / hdg_ref_set`
Heading bug value and set flag.

`alt_ref_m / alt_ref_ft / alt_ref_set`
Altitude bug values.

`vsi_ref_fpm / vsi_ref_err_fpm`
VSI bug and error.

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Green ■ marks commands new in v24. For pilot/operator interface see Operator Manual v24. Reference bugs are display-only. Observer never acts without pilot approval. Monitor never controls the aircraft.